2412ET083

Roll No. 23ugbce091013

**Course: B.Tech Computer Engineering /**
**B.Tech COMPUTER SCIENCE AND ENGINEERING**
**(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**
**Subject: Artificial Intelligence**
**Subject Code: ETCS-209**
**Semester: III**

**Time: 03 Hours** **Max Marks: 70**

**Instructions to the Students:**

1. This Question paper consists of two Sections. All sections are compulsory.
2. Section A comprises 10 questions of short answer type. All questions are compulsory. Each question carries 2 marks.
3. Section B comprises 8 long answer type questions out of which students must attempt any 5. Each question carries 10 marks.
4. Do not write anything on the question paper.

| Q.No. | SECTION –A (SHORT ANSWER TYPE QUESTIONS) | Marks |
|---|---|---|
| 1. a. | Explain state space representation with an example. | (2) |
| b. | Describe Graph and Tree data structure. How we can utilize them in AI. Explain it with examples. | (2) |
| c. | Describe key technology changes that happened during our history leading to the Artificial Intelligence era. | (2) |
| d. | Write a PROLOG code to demonstrate a stack data structure. | (2) |
| e. | Explain the components of a game search algorithm with examples. | (2) |
| f. | Define temporal models with an example. | (2) |
| g. | What do you understand about Knowledge Representation and Reasoning? What are its applications in our lives? | (2) |
| h. | Differentiate between semantic networks and frames with examples. | (2) |
| i. | Explain Logic-based Reasoning Systems with an example. | (2) |
| j. | Explain the difference between search with open and closed list using an example. | (2) |

## SECTION –B (LONG ANSWER TYPE QUESTIONS)

2. Explain the AI agent and its types with a diagram and example? Write a program in PROLOG to implement a utility agent. (10)

3. Write a program in PROLOG to implement Best First search and A* algorithm using a problem statement of your choice. (10)

4. Explain the components of heuristic search algorithms. How can we calculate a heuristic function? Write a sample PROLOG code to illustrate heuristic search. (10)

5. Explain Bayesian Networks. Explain its construction and inference mechanism. Write a program in PROLOG to implement it. (10)

6. a) Describe Bayes theorem with an example. Consider the problem statement as below: (5)

P(A) = The prior probability of winning the game= 0.05%

P(B|A) = Probability of winning the game given that you have won= 0.90

P(B|¬A) = The probability of winning the game given that you have not won = 0.05

Consider that you play the game and win. Calculate the probability that you actually have won the game, P(A|B) ?

b) Explain Hidden Markov model (HMM). Write a program in PROLOG to implement HMM. (5)

7. What is a Knowledge Base? Explain the main components of its construction and write a program in PROLOG to implement it. (10)

8. a) Define each component of First Order Logic (FOL) with a PROLOG code. (5)

b) Explain how a Cut works in PROLOG with an example code. (5)

9. c) Write a program in PROLOG to implement trigonometric functions. (5)

d) Write a program in PROLOG to implement factorial (N, F) where F represents the factorial of a number N. (5)

===END OF PAPER===